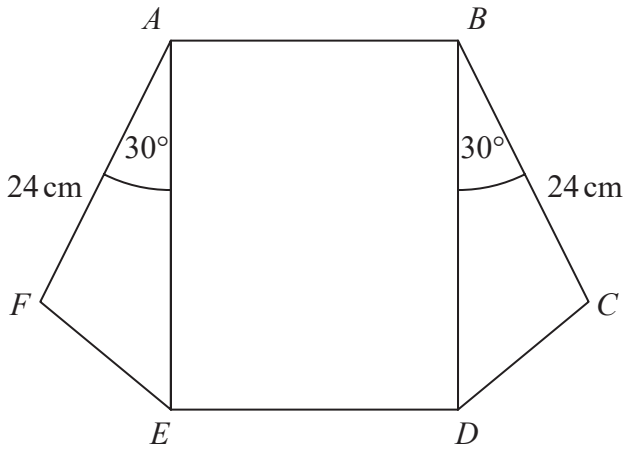
14 The diagram shows a rectangle, *ABDE*, and two congruent triangles, *AFE* and *BCD*.



area of rectangle *ABDE* = area of triangle *AFE* + area of triangle *BCD*

$$AB : AE = 1 : 3$$

Work out the length of *AE*.

sin 30° = 1/2. Area each triangle:

½ × 24 × AE × 1/2 = 6AE.

1 mark
Triangle area = 6AE.

Two triangles = 12AE.

1 mark
Two triangles = rectangle.

Rectangle AB × AE = 12AE; AB = 12.

1 mark
Find AB = 12.

AB : AE = 1 : 3 → AE = 36 cm.

1 mark
AE = 36 cm.

36 cm

(Total for Question 14 is 4 marks)